

A Duplication Analysis-Based Evolutionary Algorithm for Biobjective Feature Selection

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Abstract—Feature selection is a complex optimization problem with important real-world applications. Normally, its main target is to reduce the dimensionality of the dataset and increase the effectiveness of the classification. Owing to the population-inspired characteristics, different evolutionary algorithms (EAs) have been proposed to solve feature selection problems over the past decades. However, the majority of them only consider single-objective optimization while many real-world problems have multiple objectives, which creates a genuine demand for designing more suitable and effective EAs to handle multiobjective feature selection. A multiobjective feature selection problem usually consists of two objectives: one is to minimize the number of selected features and the other is to minimize the error of classification. In this article, we propose a duplication analysis-based EA (DAEA) for biobjective feature selection in classification. In the proposed algorithm, we make improvements on the basic dominance-based EA framework in three aspects: first, the reproduction process is modified to improve the quality of offspring; second, a duplication analysis method is proposed to filter out the redundant solutions; and third, a diversity-based selection method is adopted to further select the reserved solutions. In the experiments, we have compared the proposed algorithm with five state-of-the-art multiobjective EAs (MOEAs) and tested them on 20 classification datasets, using two widely used performance metrics. According to the empirical results, DAEA performs the best on most datasets, indicating that DAEA not only gains outstanding optimization performance but also obtains good classification and generalization results.

Index Terms—Classification, duplication analysis, evolutionary algorithm (EA), feature selection, multiobjective optimization.

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I. INTRODUCTION

FEATURE selection in classification [1], [2] is a hot research topic in machine learning, which aims to increase the classification accuracy by selecting only a small subset of relevant features from the original large set of features [3], [4]. A multiobjective feature selection problem [5]–[8] often holds two main objectives: 1) reducing the number of features and 2) improving the classification performance, which are potentially conflicting with each other [9]. Some recent works related to feature selection also study the so-called cost-based feature selection, due to the need of minimizing the computational cost associated with features [10]–[12]. In the past decades, many metaheuristics [13]–[17], including the evolutionary algorithms (EAs [18], [19]), have been proposed to solve feature selection [20]–[22] and many other real-world optimization problems [23]–[29]. Compared with traditional search methods, EAs need no domain knowledge and assumption about the search space. Moreover, their population-based search mechanism can produce a set of nondominated solutions with the tradeoff between different conflicting objectives in a single run, which is particularly suitable for multiobjective optimization problems (MOPs). Such EAs can also be called multiobjective EAs (MOEAs) [30]–[32].

Nowadays, the mainstream of MOEAs can be roughly divided into four categories based on the environmental selection strategies. The first category of MOEAs is based on the Pareto dominance [33]–[41], which analyzes the dominance relation among solutions in the objective space and ranks solutions into different nondominated fronts by nondominated sorting. The second category is based on the decomposition approach [42]–[53], which decomposes a multiobjective problem into a series of subproblems, respectively, related to a distinct and exclusive direction vector. The third category of MOEAs is based on the performance indicator [54]–[64], which adopts a certain performance metric to estimate the fitness of solutions. The fourth category denotes all the other kinds of MOEAs, such as the decision variable analysis-based EAs [65], [66], the distance measure-based EAs [67], [68], the surrogate-assisted [69] or also known as data-driven [70], [71] EAs, and the cooperative coevolutionary algorithms [72], [73].

The direction vectors and the decomposition approaches make it easier for the decomposition-based EAs to control the balance between the diversity and the convergence. However, they highly rely on the setting of the decomposition parameters and the distribution of the direction vectors. As for the indicator-based EAs, they are highly relevant to the performance metric chosen to evaluate the fitness, which can

be computationally expensive and have their own search preferences. In contrast, the nondominated sorting method used in most dominance-based EAs to provide the convergence pressure is Pareto compliant, which requires no extra parameter and less computational resource. Moreover, its largest disadvantage, difficult to converge in many-objective search space, does not exist in biobjective feature selection. However, the diversity maintenance and the reproduction process in most dominance-based EAs are not so effective to solve the biobjective feature selection problem, which need to be improved.

Reproducing high-quality offspring is also one of the major factors affecting both diversity and convergence abilities of an EA. There are already a number of works studying the reproduction process in the particle swarm optimization (PSO) or differential evolution-based multiobjective evolutionary feature selection, such as the one-bit purifying search strategy utilizing the importance degree of features [74], and the local search mechanisms based on differential learning [75], and based on single feature performance ranking [8]. However, most studies adopt similar ideas as elite preserving or local searching, which may trap the population into local optima and deteriorate the exploration diversity. Besides, there are still few studies about the reproduction process in genetic algorithms (GAs [76], [77])-based multiobjective evolutionary feature selection, despite the fact that GA is one of the mainstream EAs.

Another important issue of feature selection is the frequently occurred duplication in either decision or objective space, during evolution. In fact, duplication happened in the decision space, named *decision vector duplication*, can be easily solved by removing all the duplication and remaining only one unique solution, because, obviously, it is worthless keeping multiple same solutions. However, duplication happened in the objective space, named *objective vector duplication*, have more complex influences on the optimization and classification performance, which are much harder to deal with. This is because two solutions, even with the same classification performances (i.e., objective values) on the training data, may achieve quite different performances on the test data, because the test data is kept unknown during training [78].

To solve the aforementioned issues, we propose a duplication analysis-based EA (DAEA) for the biobjective feature selection in classification, where the major contributions are summarized as follows.

- 1) In the beginning of selection, a duplication analysis method is proposed to judge which duplicated solutions in the objective space should be filtered out. The main idea is using the distances among solutions in the decision space to estimate their dissimilarities, while those with lower dissimilarities are more likely to be removed. In this way, the population diversity in the objective space is well maintained, while the potentially optimal solutions on the test data are more likely to be reserved.
- 2) After handling duplication, a diversity-based selection method is proposed to further choose solutions after nondominated sorting [79]. Unlike the traditional crowding distance method [33], which disregards the nature

of feature selection, the proposed method considers the number of solutions with the same size of selected feature subset as the first selection criterion, and combines the crowding distance as a supplementary criterion.

- 3) The reproduction process is also improved, with a niche mating method to assemble neighboring solutions in the objective space, a k -bit crossover method to avoid reproducing the same offspring as parents, and a hybrid mutation method to balance the variation rates for both the selected and unselected features.

The remainder of this article is organized as follows. First, the related work and major motivations are introduced in Section II. Then, the essential components of DAEA are, respectively, illustrated in Section III. Afterward, the experimental setups are shown in Section IV, while the empirical performances are analyzed in Section V. Finally, the conclusion and future work are provided in Section VI.

II. BACKGROUND

A. Related Work

An MOP with M simultaneously optimized objectives can be formally defined as follows:

$$\begin{aligned} &\text{minimize } \mathbf{F}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_M(\mathbf{x}))^T \\ &\text{subject to } \mathbf{x} \in \Omega \end{aligned} \quad (1)$$

where $\mathbf{x} = (x_1, x_2, \dots, x_D)$ is a solution vector in the D -dimensional decision space Ω , $\mathbf{F}(\mathbf{x})$ is the objective vector of $\mathbf{x}$ in the M -dimensional objective space, and $f_i(\mathbf{x})$ is the i th objective function of $\mathbf{x}$. In this work, only two objectives need to be optimized: 1) the number of selected features and 2) the classification error of using the selected features. Additionally, these two objective values need to be transformed to the ratios between 0 and 1. Assuming a solution selects k features, then $f_1(\mathbf{x})$ is equal to k/D , denoting the *selected feature ratio*, while $f_2(\mathbf{x})$ denotes the *classification error rate*.

In this work, the representation of an individual is *binary coded* with genes set as either 0 or 1, while value 1 means the corresponding feature is selected and value 0 means not. In addition, according to our previous work [3], evolutionary feature selection algorithms can be roughly classified into two categories: 1) wrapper approaches and 2) filter approaches. Wrapper approaches [9], [80] include a classifier as the black box to evaluate the solution fitness (i.e., the classification performance). Filter approaches [81], on the contrary, are independent of any classifier, ignoring the classification performance of selected features. It should be noted that, in the proposed DAEA approach, only the wrapper approach is adopted for feature selection.

B. Major Motivations

EAs have already been widely applied to multiobjective feature selection [6], [9], [15], [20], [21], [80], but most studies simply apply EAs without considering the characteristics of feature selection itself [3]. One important characteristic is the solution duplication that frequently occurs in both decision and objective space of feature selection. As we know, a continuous optimization problem rarely has duplicated solutions

in either objective or decision space. However, for a discrete or combinatorial optimization problem like feature selection, solution duplication is a common and tough issue, especially when there are many redundant features or very few training instances. As mentioned before, a duplication can happen in two cases: 1) the decision vector duplication and 2) the objective vector duplication. The former can be easily handled by reserving only the unique points in the decision space. However, the latter is much more complicated to tackle due to the following reasons.

First, if all the duplicated solutions in the objective space are reserved, the population diversity in the objective space will deteriorate, due to the overlapping of objective vectors. If we consider the worst case where all the solutions map to the same objective vector, there will be no diversity at all in the objective space. Second, if only one solution is reserved for each duplicated objective vector, the population diversity in the objective space will be the best, with the maximum number of unique points. However, it is difficult to decide which duplicated solutions should be removed during optimization or training, because it is possible that these solutions duplicated in the objective space are actually unique in the decision space, and could have quite different classification accuracies on the unseen test data. The above two situations are conflicting, while removing duplicated solutions in the objective space is good for optimization but could cause deterioration on classification. Therefore, we need to find out a compromise between optimization (i.e., increasing the population diversity in the objective) and classification (i.e., reserving as many potentially optimal solutions of the test data as possible), which requires a considerate duplication analysis.

C. Further Discussion

In fact, the duplication problem focused in this article has some similarities with the multimodal MOP studied in [82]. The multimodal multiobjective feature selection allows an algorithm to reserve more than one feature subsets mapping to the same objective vector, which is actually an implicit precondition set in this article. Thus, in our experiments, all the comparison algorithms are run in the multimodal mode by default. This is quite reasonable and universal, because most traditional MOEAs actually set no restriction to select duplicated objective vectors, such as NSGA-II [33] and MOEA/D [42], i.e., two most classical MOEAs.

Despite of their similarity, they are essentially different in the following three aspects. First, in [82], the researchers treated traditional MOEAs as unimodal MOEAs, which is just contrary to this article. Second, the major motivation of [82] is to find more feature subsets with the same objective vector in order to provide not only the number of features but also the positions of the selected feature for users to make preferential decisions, while our primary goal is to filter out some redundant or less diverse feature subsets with the same objective vector in order to improve the population diversity in the objective space. Third, the algorithm framework used in [82] is based on PSO, while our proposed algorithm is based on the GA framework, which makes their algorithm designs and evolutionary mechanisms totally different.

Algorithm 1 DAEA(N, D, τ)

Input: population size N ; decision space dimensionality D ; termination criterion τ ;
Output: population Pop ;
1: **if** $(3 * N) < D$ **then**
2: randomly sample $Pop = (\mathbf{x}^1, \dots, \mathbf{x}^N)$ from the decision space
 s.t. $\sum_j \mathbf{x}_j^i \leq (3 * N), j = 1, \dots, D$;
3: **else**
4: randomly sample $Pop = (\mathbf{x}^1, \dots, \mathbf{x}^N)$ from the decision space
 s.t. $\sum_j \mathbf{x}_j^i \leq D, j = 1, \dots, D$;
5: **end if**
6: set the objective function evaluation number $t = N$;
7: **while** $t < \tau$ **do**
8: run $S = \text{Reproduction}(Pop)$ in **Algorithm 4**;
9: $t = t + N$;
10: get union population $Pop = Pop \cup S$;
11: remove all the duplicated solutions in decision space;
12: run $Pop = \text{Dup_Analysis}(Pop)$ in **Algorithm 2**;
13: rank Pop into different fronts via nondominated sorting;
14: find k **s.t.** $\text{num}(k) \geq |Pop|$ & $\text{num}(k-1) < |Pop|$;
15: set α and β as the indexes of the first $(k-1)$ fronts' and the k th front's solutions respectively;
16: run $Pop = \text{Div_Select}(Pop, \alpha, \beta, N)$ in **Algorithm 3**;
17: **end while**

III. PROPOSED ALGORITHM

In this section, we first provide the general framework of DAEA, and then, respectively, illustrate three essential components of DAEA, corresponding to the previously introduced three major contributions of this work.

A. General Framework

As shown in Algorithm 1, the general framework of DAEA can be divided into three parts: 1) initialization; 2) reproduction; and 3) environmental selection. The initialization mechanism of DAEA is actually optimized for large-scale feature selection. As shown in line 4, DAEA still adopts the traditional initialization mechanism that randomly selects features without any constraint from the decision space, when the total number of features is no more than three times larger than the population size. However, when the number of features becomes much larger, DAEA will randomly sample initial solutions from the decision space with a constraint that the number of selected features for each solution is not beyond three times the population size, as shown in line 2. There are two reasons for setting this constraint: first, according to our previous research [4], using a smaller number of features for initialization could benefit the convergence of feature selection to some extent; second, a much smaller population is hard to cover the whole decision space, so narrowing the exploration scale to three times the population size is a relatively tradeoff choice.

The termination criterion of DAEA is based on the number of objective function evaluations, i.e., (1), which increases by N each generation. In the while loop (lines 7–17), the reproduction process (line 8) will be illustrated in Section III-D, while the environmental selection process is from lines 10 to 16. The selection operation of DAEA is mainly divided

Algorithm 2 *Dup_Analysis*(*Pop*)

Input: population *Pop*;
Output: population *Pop*;

- 1: generate a $|Pop| \times |Pop|$ matrix *Dist*, while *Dist*(*i*, *j*) denotes the Manhattan distance between *Pop*(*i*) and *Pop*(*j*);
- 2: get objective vectors *Obj* of solutions in *Pop*, and find all the unique objective vectors *Obj^u*;
- 3: generate an empty removal set *R*;
- 4: **for** $i = 1, \dots, |Obj^u|$ **do**
- 5: find indexes j s.t. *Obj*(*j*) == *Obj^u*(*i*);
- 6: **if** $|j| > 1$ **then**
- 7: set *t* as the numbers of selected features and *D* as the numbers of total features;
- 8: generate a $|j|$ size dissimilarity set *Diss*;
- 9: **for** $k = 1, \dots, |j|$ **do**
- 10: *Diss*(*k*) = min(*Dist*(*j*(*k*), *j*));
- 11: **end for**
- 12: *Diss* = (*Diss*/2)/*t*;
- 13: set threshold $\delta = 0.8 - 0.6 * (t - 1)/(D - 1)$, as noted in the text below;
- 14: find indexes k s.t. *Diss*(*k*) < δ ;
- 15: $R = R \cup j(\text{randi}(k, |k| - 1))$;
- 16: **end if**
- 17: **end for**
- 18: remove solutions *Pop*(*R*) from *Pop*;

into three parts: 1) the duplication analysis (line 12); 2) the nondominated sorting (lines 13–15); and 3) the diversity maintenance (line 16). In fact, the second part of selection has no difference from the traditional dominance-based MOEAs [33], [40], [79], while the method *num*(*k*) denotes the number of solutions in the first *k* fronts (line 14). More details of the non-dominated sorting method can be referred to [33]. Moreover, it should be noted that all the duplicated solutions in the decision space are eliminated in the beginning of environmental selection (line 11). As for the computational complexity, normally, the most time-consuming part of multiobjective evolutionary feature selection is the process of evaluating the objective values, i.e., training the classification model with a selected feature subset. Since the proposed algorithm needs no extra objective function evaluations, such as using additional local search, its computational complexity remains a reasonable level.

B. Duplication Analysis

As shown in Algorithm 2, the main idea of duplication analysis is using the Manhattan distances among solutions in the decision space to estimate their dissimilarity degrees in order to decide which duplicated solutions in the objective space should be removed. In line 5, variable *j* is a set of indexes corresponding to the solutions *subject to* (abbreviated as *s.t.* in the pseudocodes) the condition that they have the same objective vector, in order to find out all the solutions with duplicated objective vectors. In lines 8–12, the dissimilarity degree of a solution is calculated by its minimum Manhattan distance to all the other solutions in the decision space, the value of which should be between 0 and 1. The Manhattan distance between two decision vectors actually reveals how different their selected features are. In line 13, the removal

	$\delta = 0.543$								<i>Diss</i>
x^1	1	1		1			1		0.25
x^2	1			1	1		1		0.25
x^3		1				1		1	0.75
x^4	1	1	1	1					0.25

Fig. 1. Example of the proposed duplication analysis method.

threshold (δ) is set according to the number of selected features (*t*) and the number of total features (*D*). The value of δ varies from 0.2 to 0.8, as the value of *t* changes from *D* to 1, making the removal criterion more inclusive. Then, in lines 14 and 15, those duplicated solutions are separated into two groups according to their dissimilarity degrees: 1) the remote solutions with dissimilarity higher than or equal to the removal threshold and 2) the clustered solutions with dissimilarity lower than the removal threshold. Among them, all the remote solutions and only one clustered solution are reserved, while the method *randi*(*k*, $|k| - 1$) means randomly selecting ($|k| - 1$) members from the array *k* (line 15). Thus, a larger value of δ will make the duplicated solutions more likely to be removed. Additionally, because δ is set to be negatively correlated with *t* (line 13), it requires a higher dissimilarity for the solution with a smaller number of selected features to be reserved.

An example of the above-mentioned duplication analysis method is shown in Fig. 1, where four different solutions ($x^1, \dots, x^4$) for a feature selection problem with totally eight features in the decision space are assumed to have the same objective vector, mapping to the same point in the objective space. In Fig. 1, each row denotes a solution vector and each column denotes a feature, while the value 1 means the corresponding feature is selected. According to lines 8–12 in Algorithm 2, the dissimilarity *Diss* for each solution is calculated as 0.25, 0.25, 0.75, and 0.25, respectively, while the current threshold $\delta = 0.8 - 0.6 * (4 - 1)/(8 - 1)$ is approximately equal to 0.543. Thereby, according to lines 14–18 in Algorithm 2, x^3 belonging to the remote group is directly reserved, while the other three solutions belonging to the clustered group can randomly reserve one. As a result, only two of the four duplicated solutions are reserved, based on the proposed duplication analysis method.

C. Diversity Maintenance

The pseudocode of the diversity maintenance process is shown in Algorithm 3, where the input variables α and β come from lines 15 and 16 of Algorithm 1, corresponding to the indexes of the solutions in the first few nondominated fronts. As shown in Algorithm 3, the proposed diversity-based selection method adopts two selection criteria: 1) the diversity score (lines 5–11) and 2) the crowding distance (lines 2–4). The former acts as the main criterion while the latter acts as a supplement. To be more specific, the sorting in line 4 makes

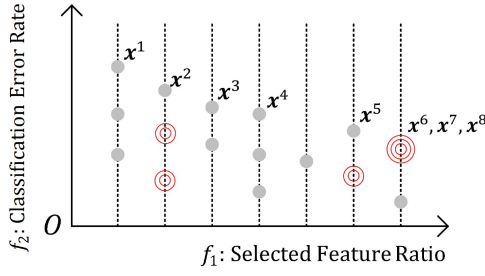


Fig. 2. Example of the proposed diversity-based selection.

Algorithm 3 *Div_Select*(Pop, α, β, N)

Input: population Pop ; selected indexes α ; unselected indexes β ; population size N ;
Output: population Pop ;

- 1: get objective vectors Obj of solutions in Pop ;
- 2: calculate crowding distances C for solutions in $Pop(\beta)$;
- 3: **while** $|\alpha| < N$ **do**
- 4: sort β according to the descend order of $C(\beta)$;
- 5: generate an empty diversity score set Div ;
- 6: **for** $i = 1, \dots, |\beta|$ **do**
- 7: find indexes j s.t. $Obj(\alpha(j), 1) == Obj(\beta, 1)$;
- 8: $Div = Div \cup |j|$;
- 9: **end for**
- 10: $k = \beta(\arg \min_i (Div(i)))$;
- 11: remove k from β but add k into α ;
- 12: **end while**
- 13: $Pop = Pop(\alpha)$;

the solution with a larger crowding distance comes first. Then, in lines 7 and 8, the diversity score of an unselected solution is set as the number of the already selected solutions with the same number of selected features, i.e., the same value of the 1st-objective f_1 that has already been defined in the first paragraph of Section II-A. Finally, the solution with the smallest diversity score is selected (lines 10 and 11), while the crowding distance only takes effect when there are multiple solutions with the same smallest diversity score, based on the sorting beforehand (line 4). Since the optimal solutions on the training data may not be the optimal solutions on the test data, i.e., there might be an overfitting issue, it is important to maintain a good diversity for each size of the selected feature subset, i.e., in the f_1 objective direction.

An example of Algorithm 3 is presented in Fig. 2, where the gray dots denote unique solutions and the red concentric circles denote duplicated solutions (one circle one duplicated solution), all in the objective space. We assume that solutions $\{x^1, x^2, \dots, x^8\}$ are in the last nondominated front (the k th front) that have not been selected (some of them will need to be selected to form the new population), while all the other solutions are in the first $k - 1$ nondominated fronts that have already been selected, by the nondominated sorting action taken in lines 13–15 of Algorithm 1. In Fig. 2, solutions with the same number of selected features lie in the same dashed vertical line, while the diversity score calculated in Algorithm 3 actually means the number of already selected solutions that have the same selected feature ratio with a certain unselected solution. Thus, the diversity score is equal to the number of already selected solutions on the same

Algorithm 4 *Reproduce*(Pop)

Input: population Pop ;
Output: offspring set S ;

- 1: set the number of offspring $N = |Pop|$;
- 2: set the neighborhood size $T = \max(4, \lceil N * 0.2 \rceil)$;
- 3: temporarily normalize the objective vectors of all the solutions in Pop to the scale $[0, 1]$;
- 4: generate a $N * T$ neighborhood matrix Nic , while $Nic(i)$ consists of T nearest solutions to $Pop(i)$, according to the Euclidean distances among normalized objective vectors;
- 5: generate an empty parent set Par ;
- 6: **for** $i = 1, \dots, N$ **do**
- 7: **if** $rand < 0.8$ **then**
- 8: randomly select a parent from $Nic(i)$ into Par ;
- 9: **else**
- 10: randomly select a parent from Pop into Par ;
- 11: **end if**
- 12: **end for**
- 13: initialize the offspring set $S = Pop$;
- 14: **for** $i = 1, \dots, N$ **do**
- 15: find indexes j s.t. $Pop(i, j) \neq Par(i, j)$;
- 16: **if** $|j| > 1$ **then**
- 17: $k = randi(j, randi(\{1, \dots, |j| - 1\}, 1))$;
- 18: do crossover: $S(i, k) = Par(i, k)$;
- 19: **end if**
- 20: **end for**
- 21: run $S = Mutate(S)$ in Algorithm 5;

dashed line, and the diversity scores for the eight solutions $\{x^1, x^2, \dots, x^8\}$ in Fig. 2 are counted as $\{2, 4, 1, 2, 2, 1, 1, 1\}$, respectively. For instance, there are two selected solutions which have the same number of features as solution x^1 , therefore, the diversity score of x^1 is 2. Obviously, x^3, x^6, x^7 , and x^8 all rank the first by diversity score. Since there are multiple choices, we further compare their crowding distances, turning out that x^6, x^7 , and x^8 are superior to x^3 with larger crowding distances (the crowding distance of each boundary solution is automatically set to infinite according to [33]). However, the crowding distances of x^6, x^7 , and x^8 are still the same because they overlap in the objective space. In this case, we just randomly select one of them, such as x^8 . It should be noted that the selection process is dynamic and only selects one solution in each round. Thereby, in the second selection round, x^3 will rank better than x^6 and x^7 by diversity score, due to the previous selection of x^8 .

D. Reproduction Modification

As shown in Algorithm 4, the reproduction process of DAEA is mainly divided into three parts: 1) the niche mating (lines 1–12); 2) the k -bit crossover (lines 13–20); and 3) the hybrid mutation (line 21). The niche mating method randomly selects parents with a proportion of 80% or 20% from the local neighborhood or the global population alternatively, which is similar to the regular settings, i.e., 85%/15% in [9] and 90%/10% in [43]. We adopt the idea of niching because it is a stabler and more effective way to exchange evolutionary information between two solutions with similar objective vectors, as already proved in many research works [9], [43], [46], [47]. In line 2, the neighborhood size is set to 20% of the population size, similar to the regular setting 10% in [46],

Algorithm 5 *Mutate*(S)

Input: offspring set S ;
Output: offspring set S ;

```

1: for  $i = 1, \dots, |S|$  do
2:   if  $\text{rand} < 0.2$  then
3:     find indexes  $t^1$  s.t.  $S(i, t^1) == 1$ ;
4:     for  $k = 1, \dots, |t^1|$  do
5:       if  $\text{rand} < 1/(1 + |t^1|)$  then
6:         do mutation:  $S(i, t^1(k)) = \tilde{S}(i, t^1(k))$ ;
7:       end if
8:     end for
9:     find indexes  $t^0$  s.t.  $S(i, t^0) == 0$ ;
10:    repeat Lines 4 to 8 but replace  $t^1$  with  $t^0$ ;
11:   else
12:     get the decision space dimension  $D = |S(1)|$ ;
13:     for  $l = 1, \dots, D$  do
14:       if  $\text{rand} < 1/D$  then
15:         do mutation:  $S(i, l) = \tilde{S}(i, l)$ ;
16:       end if
17:     end for
18:   end if
19: end for

```

and at least four, considering the left, right, up, and down four directions in the objective space. In line 3, the objective values in each objective direction are separately normalized to the same scale, for the sake of fairness on choosing neighboring solutions afterward (line 4).

In lines 13–20 of Algorithm 4, the proposed k -bit crossover method randomly modifies at least one but no more than $(|j| - 1)$ decision variables on one of the two parents [e.g., $\text{Pop}(i)$ and $\text{Par}(i)$ in line 15]. Here, DAEA only does crossover on the bits, where the decision variable values of the two parents are different (e.g., j in line 15). For example, if we take solutions $\mathbf{x}^1$ and $\mathbf{x}^3$ in Fig. 1 as two parents to make an offspring, we will only do crossover randomly on the 1st, 4th, 5th, 6th, 7th, and 8th bits (since the decision variable values are different on those bits), but ignore the 2nd and 3rd bits. In this way, the offspring produced by the k -bit crossover method is more likely to be different from their parents.

The hybrid mutation method is detailed in Algorithm 5, where a modified mutation method (lines 2–10) along with a traditional mutation method (lines 11–18) are randomly applied with a roughly set ratio of 20% and 80%, respectively [83], [84]. It should be noted that we have no attempt to find the best ratio, but the probability of using the traditional bitwise mutation method should be set relatively larger, because it provides a more general and stable variation. However, if the number of selected features is much smaller than the decision space dimensionality (i.e., D in line 12), the probability of adding new features (turning 0 to 1) will become much larger than that of removing old features (turning 1 to 0). The modified mutation mechanism, in contrast, balances the probabilities of adding and removing features by separately tackling the selected and unselected features (i.e., t^1 and t^0 in lines 3 and 9, respectively). It should be noted that t^1 and t^0 in lines 3 and 9 are not the indexes of solutions, but are the indexes of value 1 or 0 genes in a certain solution decision vector, corresponding to the selected and unselected features, respectively. Moreover, the specific mutation action

in Algorithm 5 (lines 6 and 15) is actually quite simple, which only requires to flip the specific gene whose index is denoted by $t^1(k)$ in line 6 or l in line 15, by turning either 1 to 0 or 0 to 1.

Let us take an offspring with two selected features (value 1 bits) as an example to illustrate the above point, while the decision space dimensionality is 10 and the number of unselected features (value 0 bits) is 8. Thereby, in the traditional mutation method, the probability of removing at least one selected feature is $1 - (1 - 1/10)^2 = 0.19$, while that of adding at least one unselected feature is $1 - (1 - 1/10)^8 \approx 0.57$. Obviously, with about three times the probability difference, it is more likely to add features rather than to remove features, which is not good for the variation of offspring. In contrast, in the modified mutation method, the probability of removing at least one selected feature is $1 - (1 - 1/(1 + 2))^2 \approx 0.56$, while that of adding at least one unselected feature is $1 - (1 - 1/(1 + 8))^8 \approx 0.61$. It can be seen that their probability difference is greatly narrowed, with almost the same probability, which makes it more fair to mutate both 0 and 1 decision variables, increasing the chance of refreshing selected features.

IV. EXPERIMENTAL SETUP

A. Comparison Algorithms

There are five state-of-the-art MOEAs chosen for comparisons with DAEA in this study, i.e., NSGA-II [33], MOEA/D [42], MOEA/HD [43], NSGA-II/SDR [85], and PMEA [86]. Among them, NSGA-II is the most classic dominance-based MOEA which adopts the same nondominated sorting and crowding distance calculation methods as DAEA. MOEA/D is the most classic decomposition-based MOEA whose convergence is based on the fitness of aggregation functions instead of the Pareto dominance relationship. MOEA/HD (our recently proposed work) is an advanced version of MOEA/D, which adaptively adjusts the distribution of weight vectors in order to tackle complex Pareto fronts. NSGA-II/SDR is a modified version of NSGA-II, which adopts a strengthened dominance relation to better balance convergence and diversity. PMEA (our another recently proposed work) is an indicator-based MOEA that makes use of polar-metric [87] as the environmental selection criterion.

Moreover, the variation operators used for all the above comparison algorithms are the single-point crossover and bitwise mutation methods. The crossover rate is set to 1, while the mutation rate is set to $1/D$ with D as the number of decision variables. In fact, we have also compared KnEA (the knee point-driven MOEA [67]) and HypE (the hypervolume indicator-based MOEA [55]), with DAEA, while the experimental results turn out that DAEA outperforms them on all the datasets tested in this article. However, their performances are not presented in this article, due to the page limit.

B. Classification Datasets

There are 20 datasets on classification used in the experiment, while all of them are based on the real-world data and can be downloaded from the UCI machine learning repository [88]. As shown in Table I, the number of features varies

TABLE I
CLASSIFICATION DATASETS IN ASCEND ORDER OF FN

No.	Dataset Name	Instance	Feature	Class
1	Wine	178	13	3
2	Climate	540	18	2
3	Breast_cancer_WD	569	30	2
4	Ionosphere	351	34	2
5	SPECTF_Heart	267	44	2
6	Connectionist_Bench_Sonar	208	60	2
7	Mice_Protein_Expression	1077	77	8
8	Libras_Movement	360	90	15
9	Hill_Valley	606	100	2
10	MUSK1	476	166	2
11	Semeion_Handwritten_Digit	1593	256	10
12	Arrhythmia	452	278	16
13	LSVT_Voice_Rehabilitation	126	310	2
14	Madelon_Train_Validation	2600	500	2
15	ISOLET5	1559	617	26
16	Multiple_Features	2000	649	10
17	SRBCT	83	2308	4
18	Leukemia1	72	5327	3
19	Brain1	90	5920	5
20	Leukemia	72	7070	2

from 13 (i.e., Wine) to 7070 (i.e., Leukemia), which covers both simple and hard problems. Moreover, the number of classes varies from 2 (i.e., Climate) to 26 (i.e., ISOLET5), while half of the datasets are multiclass classification problems and the other half are binary classification problems. The number of instances also varies, while the smallest one is 72 (i.e., Leukemia) and the largest is 2600 (i.e., Madelon). Therefore, the datasets used in this work are both comprehensive and complicated enough to challenge the generalization and effectiveness of each competing algorithm for tackling the feature selection in classification.

C. Performance Indicators

There are two performance indicators used in this article, i.e., hypervolume (HV) [54] and inverted generation distance (IGD) [89]. For HV, at first, all the objective values obtained by each algorithm are normalized by the ideal and nadir points to the scale $[0, 1]$ in each objective direction, while the ideal and nadir points are, respectively, set to $(0, 0)$ and $(1.1, 1.1)$. Then, the reference point of HV is set to $(1, 1)$. For IGD, unlike other benchmark test problems (like DTLZ [90] or WFG [91]), feature selection is a discrete and complex problem, which is almost impossible to calculate the exact Pareto front. Thus, for the sake of fairness, we first collect the final populations of all the competing algorithms and merge them into a union population. Then, this union population is ranked by nondominated sorting, while only the first nondominated front is reserved as the reference points of IGD. Generally speaking, the smaller the IGD values are, the better performances an algorithm gains, which is opposite for HV.

Moreover, the Wilcoxon test with a significance level of 0.05 is also applied to check whether the performance of DAEA is significantly different from that of any other competing algorithm. For relative performance rankings among all the competing algorithms, the Freidman test is also adopted. Here, we need to do a small transformation on HV values, so that

HV and IGD can be consistent in the aspect of the Freidman test, i.e., better HV and IGD performances can both correspond to smaller Freidman ranking values, making it more convenient for experimental analyses and comparisons. Thus, if hv denotes the normalized HV value between 0 and 1, we will use $(1 - hv)$ for the Freidman test, instead of directly using hv .

D. Parameter Settings

In this work, all the experiments are coded in MATLAB, based on an open-source platform, PlatEMO [92]. Each competing algorithm is run for 30 times independently on each dataset, while the initial random seed is set the same for all the algorithms but changes by different datasets and runs. Thus, all the performances can be reproduced anytime and anywhere by other researchers. The specific parameters for all the chosen algorithms are referred to [33], [42], [43], [55], and [67]. The population size N is adaptively set to the number of features but not beyond 200 in order to balance the diversity and efficiency, which means N for those datasets with feature numbers (FNs) larger than 200 will still be set to 200. The maximum objective function evaluation number, which acts as the termination criterion for each algorithm, is set to 100 times of N for all the datasets, meaning that the maximum number of evolutionary generations is about 100.

As for classification, each dataset is randomly divided into a training subset and a test subset with the proportions of nearly 70% and 30%, respectively, which also follows a stratified split process [3], [4], [8]. It should be noted that this division is fixed for a specific dataset on all the 30 runs by each algorithm so that the statistical performance makes sense, but varies by different datasets. During the training process, KNN with tenfold cross-validation is applied to calculate the classification error on the training set to avoid the feature selection bias [78], [93], while the number of nearest neighbors in KNN is set to 5 so as to balance the accuracy and efficiency.

V. EMPIRICAL PERFORMANCE

A. Overall Empirical Result Analysis

In the experiment, the candidate solutions used to measure the performance of an algorithm should only rely on the information from the training data, because the test data is always kept unseen from the decision maker. Here, for a more comprehensive analysis, we use two criteria to separately select the candidate solutions on the training data. In the first criterion, only the first nondominated Pareto front from the training data is chosen for measuring the performance on both the training and test data. In the second criterion, we select solutions with the best classification accuracy for each size of the selected feature subset based on the training data. If there exist multiple best solutions for a certain number of selected features, we randomly choose one of them. It should be noted that, no matter which criterion we use, only the nondominated ones from the candidate solutions are used to further calculate the HV and IGD values. Thus, the two criteria actually obtain the same HV or IGD performance on the training data, but probably have different performances on the test data. For all

TABLE II

MEAN HV PERFORMANCE ON TEST DATA, WHILE THE BEST PERFORMANCE IS HIGHLIGHTED IN GRAY AND ANY PERFORMANCE NOT SIGNIFICANTLY DIFFERENT FROM THAT OF DAEA IS PREFIXED BY †

No.	DAEA	NSGA-II	MOEA/D	MOEA/HD	NSGA-II/SDR	PMEA
1	8.7364e-01 ± 1.52e-02	†8.7067e-01 ± 1.78e-02	†8.7349e-01 ± 1.33e-02	†8.6886e-01 ± 1.89e-02	8.2282e-01 ± 1.79e-02	†8.6999e-01 ± 1.88e-02
2	8.9084e-01 ± 7.11e-03	†8.8953e-01 ± 6.99e-03	8.8732e-01 ± 6.83e-03	†8.8958e-01 ± 7.79e-03	†8.8875e-01 ± 7.77e-03	†8.9028e-01 ± 7.18e-03
3	9.3880e-01 ± 4.59e-03	†9.3613e-01 ± 6.44e-03	9.3567e-01 ± 5.35e-03	†9.3653e-01 ± 4.02e-03	9.3197e-01 ± 1.83e-03	9.3603e-01 ± 3.65e-03
4	8.8436e-01 ± 8.83e-03	†8.8426e-01 ± 1.10e-02	†8.7955e-01 ± 1.38e-02	†8.8313e-01 ± 1.28e-02	†8.7963e-01 ± 1.31e-02	†8.8322e-01 ± 1.26e-02
5	7.9258e-01 ± 1.96e-02	†7.8895e-01 ± 2.19e-02	7.7998e-01 ± 2.16e-02	†7.9497e-01 ± 1.82e-02	†7.9757e-01 ± 2.08e-02	†7.8542e-01 ± 2.39e-02
6	8.0493e-01 ± 2.22e-02	†7.9604e-01 ± 2.47e-02	7.9275e-01 ± 2.09e-02	7.9098e-01 ± 2.21e-02	7.8907e-01 ± 2.32e-02	7.6168e-01 ± 3.83e-02
7	7.3839e-01 ± 1.44e-02	7.2116e-01 ± 2.65e-02	†7.3157e-01 ± 1.94e-02	5.1997e-01 ± 2.48e-01	2.1214e-01 ± 7.38e-02	1.8973e-01 ± 1.08e-01
8	7.5487e-01 ± 1.53e-02	7.4269e-01 ± 1.38e-02	†7.4967e-01 ± 1.83e-02	†7.6168e-01 ± 1.39e-02	†7.5061e-01 ± 1.68e-02	7.2792e-01 ± 1.79e-02
9	6.3221e-01 ± 1.10e-02	5.9437e-01 ± 2.51e-02	†6.2554e-01 ± 8.97e-03	†6.2435e-01 ± 1.72e-02	6.0534e-01 ± 2.34e-02	5.8535e-01 ± 2.10e-02
10	9.1176e-01 ± 1.19e-02	8.3703e-01 ± 2.63e-02	8.7059e-01 ± 2.16e-02	8.7141e-01 ± 1.79e-02	8.3819e-01 ± 2.80e-02	8.3437e-01 ± 1.27e-02
11	8.3136e-01 ± 9.54e-03	7.1894e-01 ± 1.25e-02	7.9948e-01 ± 1.01e-02	7.5048e-01 ± 1.24e-02	7.1336e-01 ± 1.63e-02	7.1078e-01 ± 1.22e-02
12	6.9761e-01 ± 1.34e-02	6.5983e-01 ± 1.94e-02	6.8286e-01 ± 1.18e-02	4.9876e-01 ± 1.92e-02	4.4387e-01 ± 2.39e-02	5.5338e-01 ± 6.15e-02
13	9.2841e-01 ± 2.07e-02	8.3162e-01 ± 3.19e-02	8.9962e-01 ± 2.37e-02	8.5214e-01 ± 3.32e-02	8.0134e-01 ± 2.17e-02	7.9704e-01 ± 2.55e-02
14	8.6580e-01 ± 1.71e-02	5.8170e-01 ± 1.47e-02	8.2269e-01 ± 3.38e-02	5.6826e-01 ± 1.73e-02	5.5723e-01 ± 1.69e-02	5.6539e-01 ± 1.63e-02
15	8.7795e-01 ± 8.11e-03	6.9466e-01 ± 1.48e-02	8.0031e-01 ± 1.40e-02	6.8786e-01 ± 1.30e-02	6.7477e-01 ± 1.24e-02	6.7531e-01 ± 1.09e-02
16	9.8291e-01 ± 2.24e-03	7.8710e-01 ± 9.65e-03	9.3305e-01 ± 5.61e-03	7.8347e-01 ± 1.13e-02	7.4360e-01 ± 1.52e-02	7.7089e-01 ± 9.67e-03
17	9.4363e-01 ± 3.54e-02	2.9477e-01 ± 2.00e-03	3.5521e-01 ± 1.94e-03	2.5858e-01 ± 1.71e-03	2.5882e-01 ± 2.51e-03	2.8592e-01 ± 2.36e-03
18	9.6124e-01 ± 2.64e-02	5.4235e-01 ± 1.98e-02	5.9347e-01 ± 2.61e-02	5.1841e-01 ± 2.20e-02	5.1666e-01 ± 1.69e-02	5.3994e-01 ± 1.68e-02
19	7.8884e-01 ± 4.75e-02	4.8159e-01 ± 3.09e-03	5.3751e-01 ± 4.65e-03	4.6325e-01 ± 8.93e-03	4.5613e-01 ± 3.35e-03	4.8284e-01 ± 5.93e-03
20	9.4615e-01 ± 4.75e-02	5.4403e-01 ± 1.57e-02	5.8426e-01 ± 2.95e-02	5.2332e-01 ± 1.39e-02	5.2167e-01 ± 1.92e-02	5.3950e-01 ± 1.55e-02

the tables in this article on the test data, we use *Test* to denote the performance measured by the first criterion and *Test** to denote the performance measured by the second criterion.

The detailed HV and IGD performances on each test dataset measured by the first criterion, including the mean and standard deviation metric values obtained by each algorithm, are shown in Tables II and III, respectively, while those measured by the second criterion are shown in Tables SII and SIV in the supplementary material. In addition, their detailed HV and IGD performances on each training dataset are shown in Tables SI and SIII, respectively, in the supplementary material. For the readers' convenience, we also calculate the general winning ratios for DAEA against all the other algorithms in Table IV and the average Freidman's rankings in Table V, in terms of both HV and IGD metrics.

The results clearly show that DAEA performs the best in all the training datasets and most of the test datasets. As shown in Tables II and III, the only five losses of DAEA, although with no significant difference in four of them, all happen in the relatively low-dimensional datasets. The single significant loss of DAEA happens against MOEA/HD in Table III on the Libras_Movement (the eighth) test dataset, but DAEA still takes the second best on this dataset. In fact, the performance measured by the second criterion, as shown in Table IV (Test* rows) and Tables SII and SIV in the supplementary material, is

TABLE III

MEAN IGD PERFORMANCE ON TEST DATA, WHILE THE BEST PERFORMANCE IS HIGHLIGHTED IN GRAY AND ANY PERFORMANCE NOT SIGNIFICANTLY DIFFERENT FROM THAT OF DAEA IS PREFIXED BY †

No.	DAEA	NSGA-II	MOEA/D	MOEA/HD	NSGA-II/SDR	PMEA
1	7.8094e-02 ± 1.50e-02	†7.8128e-02 ± 1.69e-02	†8.2347e-02 ± 2.22e-02	†8.0345e-02 ± 1.97e-02	1.3030e-01 ± 1.89e-02	†8.2870e-02 ± 1.84e-02
2	6.0219e-02 ± 2.59e-02	†6.4453e-02 ± 2.73e-02	†6.7568e-02 ± 3.46e-02	†6.8890e-02 ± 3.45e-02	†7.2799e-02 ± 3.46e-02	†6.5271e-02 ± 3.47e-02
3	1.9682e-02 ± 4.65e-03	†2.1808e-02 ± 5.42e-03	2.3196e-02 ± 5.12e-03	2.2723e-02 ± 5.49e-03	†2.0178e-02 ± 3.71e-03	2.3982e-02 ± 4.65e-03
4	3.9794e-02 ± 6.72e-03	†3.9482e-02 ± 1.08e-02	†4.5186e-02 ± 1.44e-02	†4.0901e-02 ± 1.01e-02	†4.5559e-02 ± 1.25e-02	†4.2245e-02 ± 1.10e-02
5	6.9643e-02 ± 1.54e-02	†7.4192e-02 ± 1.98e-02	8.2688e-02 ± 1.78e-02	†6.7355e-02 ± 1.60e-02	†6.8233e-02 ± 1.92e-02	†7.4085e-02 ± 1.67e-02
6	6.0506e-02 ± 1.43e-02	†7.0285e-02 ± 2.34e-02	†7.0021e-02 ± 1.75e-02	†7.1393e-02 ± 2.16e-02	7.1577e-02 ± 2.13e-02	8.9645e-02 ± 2.82e-02
7	2.4235e-02 ± 4.76e-03	†2.8445e-02 ± 1.26e-02	†2.8662e-02 ± 9.48e-03	2.0448e-01 ± 2.08e-01	4.5773e-01 ± 6.80e-02	4.9015e-01 ± 8.87e-02
8	5.5397e-02 ± 6.67e-03	1.1558e-01 ± 4.78e-02	†5.6363e-02 ± 1.45e-02	4.9854e-02 ± 8.77e-03	7.8715e-02 ± 3.37e-02	1.4892e-01 ± 3.04e-02
9	3.6364e-02 ± 6.39e-03	6.1550e-02 ± 2.13e-02	4.1539e-02 ± 6.75e-03	†4.0886e-02 ± 1.31e-02	5.3902e-02 ± 2.16e-02	6.6853e-02 ± 1.75e-02
10	4.2628e-02 ± 7.58e-03	8.3800e-02 ± 1.48e-02	6.4610e-02 ± 1.46e-02	6.2986e-02 ± 1.11e-02	8.1655e-02 ± 1.59e-02	8.4167e-02 ± 9.42e-03
11	4.1214e-02 ± 1.47e-02	1.6426e-01 ± 1.53e-02	7.9734e-02 ± 1.12e-02	1.3502e-01 ± 1.33e-02	1.7123e-01 ± 1.68e-02	1.7530e-01 ± 1.50e-02
12	2.3653e-02 ± 7.58e-03	5.0078e-02 ± 1.44e-02	3.3306e-02 ± 8.23e-03	1.9155e-01 ± 1.47e-02	2.7550e-01 ± 2.34e-02	1.4508e-01 ± 5.44e-02
13	4.6706e-02 ± 1.24e-02	1.1787e-01 ± 2.14e-02	6.1121e-02 ± 1.52e-02	1.0251e-01 ± 2.28e-02	1.5492e-01 ± 1.91e-02	1.4518e-01 ± 1.74e-02
14	2.4644e-02 ± 1.21e-02	2.8132e-01 ± 1.67e-02	5.8601e-02 ± 3.02e-02	3.0359e-01 ± 1.89e-02	3.0800e-01 ± 1.83e-02	2.9748e-01 ± 1.61e-02
15	1.5410e-02 ± 2.81e-03	2.4698e-01 ± 1.58e-02	1.3173e-01 ± 6.60e-03	2.5503e-01 ± 1.41e-02	2.6697e-01 ± 1.16e-02	2.6984e-01 ± 1.18e-02
16	9.7484e-03 ± 5.26e-03	2.3260e-01 ± 1.01e-02	8.1857e-02 ± 4.95e-03	2.3652e-01 ± 1.16e-02	2.7846e-01 ± 1.66e-02	2.5060e-01 ± 9.65e-03
17	4.6463e-02 ± 1.77e-02	6.0649e-01 ± 2.91e-03	5.3490e-01 ± 1.68e-03	6.6388e-01 ± 2.91e-03	6.6347e-01 ± 4.25e-03	6.1969e-01 ± 3.59e-03
18	4.0161e-02 ± 2.04e-02	4.2056e-01 ± 1.00e-02	3.4678e-01 ± 1.43e-02	4.4324e-01 ± 1.13e-02	4.4619e-01 ± 8.60e-03	4.2075e-01 ± 1.01e-02
19	7.5387e-02 ± 3.64e-02	4.1872e-01 ± 4.32e-03	3.4111e-01 ± 6.41e-03	4.4439e-01 ± 1.25e-02	4.5438e-01 ± 4.71e-03	4.1775e-01 ± 5.28e-03
20	5.9091e-02 ± 5.22e-02	4.3824e-01 ± 9.73e-03	3.8949e-01 ± 3.02e-02	4.6291e-01 ± 9.53e-03	4.6814e-01 ± 1.21e-02	4.4101e-01 ± 8.91e-03

TABLE IV

WIN, LOSS, AND DRAW RATIOS FOR DAEA AGAINST ALL THE OTHER ALGORITHMS ON ALL DATASETS, WHILE VALUES IN BRACKET DENOTE SIGNIFICANT RESULTS OR NO SIGNIFICANT DIFFERENCES

Metric	Data	Win	Loss	Draw
HV	Train	100% (94%)	0% (0%)	0% (6%)
	Test	97% (74%)	3% (0%)	0% (26%)
	Test*	98% (89%)	2% (0%)	0% (11%)
IGD	Train	100% (95%)	0% (0%)	0 (5%)
	Test	96% (72%)	4% (1%)	0 (27%)
	Test*	98% (82%)	2% (0)	0 (18%)

TABLE V

AVERAGE FREIDMAN'S RANKINGS ON ALL DATASETS

Metric	Data	DAEA	NSGA-II	MOEA/D	MOEA/HD	NSGA-II/SDR	PMEA
HV	Train	1.24	3.54	3.03	3.78	5.03	4.38
	Test	1.79	3.60	2.68	3.73	4.79	4.42
	Test*	1.60	3.72	2.77	3.75	4.66	4.49
IGD	Train	1.33	3.51	3.01	3.89	5.06	4.20
	Test	1.81	3.55	2.65	3.82	4.75	4.41
	Test*	1.68	3.64	2.71	3.82	4.70	4.46

even more outstanding, with a greater ratio of significant wins and a bit smaller ratio of losses. This is because the second criterion used to select candidate solutions is more consistent with the primary goal of the proposed diversity-based selection method (i.e., Algorithm 3), which attempts to find as many

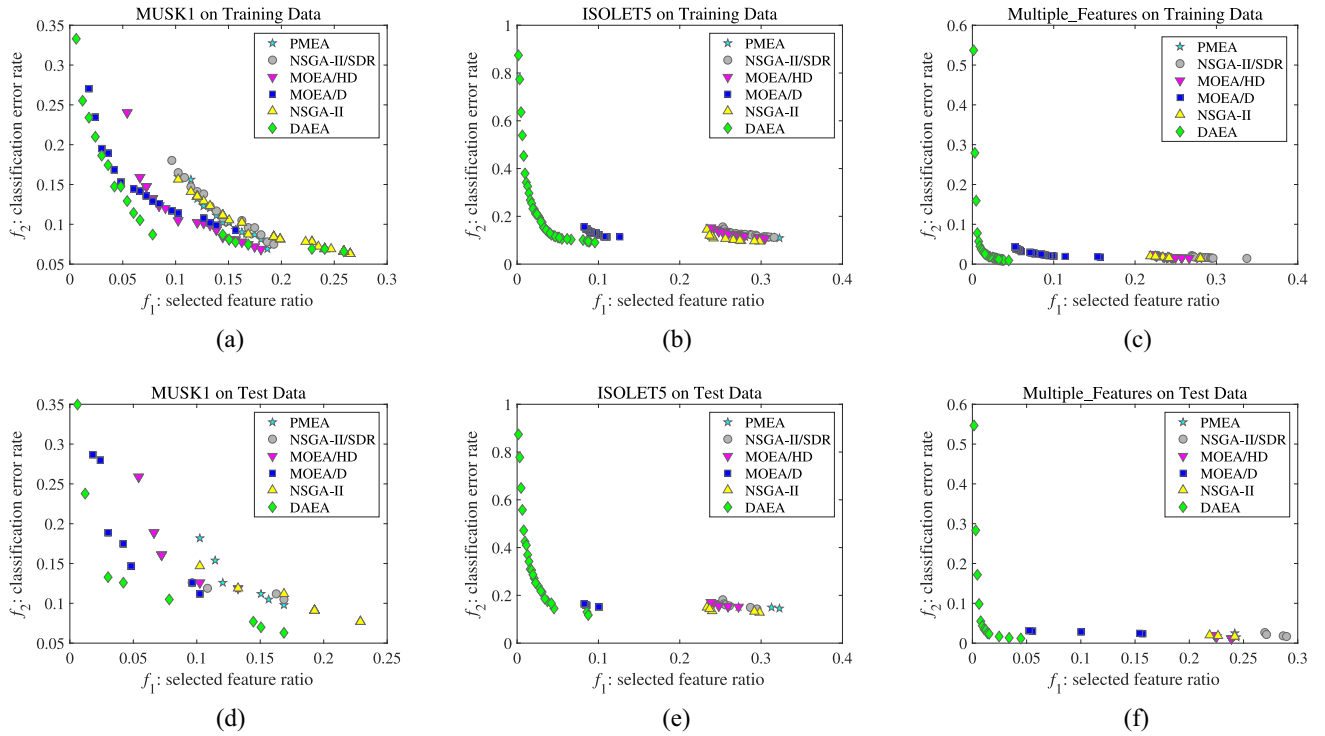


Fig. 3. Distributions of the first nondominated fronts obtained by each algorithm in the objective space. (a) MUSK1 on training data. (b) ISOLET5 on training data. (c) Multiple_Features on training data. (d) MUSK1 on test data. (e) ISOLET5 on test data. (f) Multiple_Features on test data.

potentially optimal solutions on the test data as possible for each size of the selected feature subset during training.

B. Nondominated Front Distribution Analysis

For an intuitive analysis, we also illustrate the distributions of the first nondominated fronts with median HV performance obtained by each algorithm in Fig. 3. The first row of Fig. 3 shows all the nondominated solutions on the training data, while the second row demonstrates the nondominated solutions on the test data. It should be noted that all the solutions shown in the second row are derived from the first row but removing those dominated on the test data, although they may be nondominated on the training data. We choose the three datasets (i.e., the ISOLET5, LSVT_voice, and Multiple_Features) because their nondominated front distributions obtained by different algorithms are easier to be distinguished visually. It is suggested that the nondominated solutions of DAEA are generally more diverse (i.e., reserving more Pareto-optimal solutions) and converge better (i.e., with less selected features and lower classification errors) than those of any other algorithm on both the training and test data. It is also suggested that some nondominated solutions in the first row of Fig. 3 have changed their positions or even disappeared in the second row, i.e., having different classification errors or being dominated by other solutions, which is quite common in classification problems including feature selection, mainly due to the differences between training and test data.

C. Further Feature Selection Analysis

For a further feature selection analysis, we use a new performance metric called minimal classification error (MCE),

which is exactly equal to the minimal value obtained in the second objective direction (i.e., the classification error rate). Unlike HV and IGD which also take the population diversity into consideration, MCE only focuses on the classification error of an algorithm, and prefers smaller values. Accompanied by MCE, another performance metric called FN is also adopted, which is exactly equal to the number of features selected by the solution with the MCE, i.e., the same solution associated with MCE. Different from single-objective evolutionary feature selection that chooses only one best solution on the training data, multiobjective evolutionary feature selection normally chooses all the solutions in the first nondominated front. Thus, we use the solution with the MCE among them, which is probably the most valuable solution for a decision maker, to represent the classification performance.

Tables VI and VII, respectively, provide the detailed MCE and FN performances on the test data measured by the first criterion, including the mean and standard deviation metric values obtained by each algorithm, while those measured by the second criterion are given in Tables SVI and SVIII in the supplementary material. Their MCE and FN performances on the training data are, respectively, shown in Tables SV and SVII in the supplementary material. For the readers' convenience, we also calculate the general winning ratios for DAEA against all the other algorithms in Table VIII and the average Friedman's rankings in Table IX, in terms of both MCE and FN metrics.

The results clearly show that DAEA performs the best in all the training datasets and most of the test datasets in terms of the MCE metric, which is similar to the previously analyzed HV and IGD performances. As shown in Table VI,

TABLE VI

MEAN MCE PERFORMANCE ON TEST DATA, WHILE THE BEST PERFORMANCE IS HIGHLIGHTED IN GRAY AND ANY PERFORMANCE NOT SIGNIFICANTLY DIFFERENT FROM THAT OF DAEA IS PREFIXED BY †

No.	DAEA	NSGA-II	MOEA/D	MOEA/HD	NSGA-II/SDR	PMEA
1	1.9497e-02 ± 2.55e-02	†2.6415e-02 ± 2.96e-02	†2.3270e-02 ± 1.62e-02	†2.8931e-02 ± 2.88e-02	1.0755e-01 ± 2.48e-02	†2.4528e-02 ± 3.10e-02
2	6.3580e-02 ± 1.06e-02	†6.5432e-02 ± 1.03e-02	†6.8107e-02 ± 1.01e-02	†6.5638e-02 ± 1.15e-02	†6.6872e-02 ± 1.15e-02	†6.4609e-02 ± 1.05e-02
3	3.1384e-02 ± 5.84e-03	†3.3528e-02 ± 4.59e-03	3.5088e-02 ± 6.69e-03	†3.4113e-02 ± 4.88e-03	4.0546e-02 ± 2.14e-03	†3.4503e-02 ± 4.70e-03
4	9.6855e-02 ± 1.05e-02	†9.7170e-02 ± 1.29e-02	1.0283e-01 ± 1.61e-02	†9.8428e-02 ± 1.52e-02	†1.0283e-01 ± 1.57e-02	†9.7170e-02 ± 1.45e-02
5	2.0535e-01 ± 2.39e-02	†2.0823e-01 ± 2.42e-02	2.2058e-01 ± 2.51e-02	†2.0370e-01 ± 2.12e-02	†2.0000e-01 ± 2.43e-02	†2.0082e-01 ± 2.15e-02
6	1.9516e-01 ± 2.72e-02	†1.9462e-01 ± 2.84e-02	2.0914e-01 ± 2.49e-02	2.1075e-01 ± 2.57e-02	2.1237e-01 ± 2.61e-02	2.1774e-01 ± 3.81e-02
7	2.5143e-01 ± 1.88e-02	2.7393e-01 ± 2.32e-02	†2.5808e-01 ± 2.32e-02	4.9969e-01 ± 2.86e-01	8.4274e-01 ± 8.90e-02	8.4059e-01 ± 1.17e-01
8	2.4349e-01 ± 1.85e-02	†2.3556e-01 ± 2.08e-02	†2.4921e-01 ± 2.15e-02	†2.3524e-01 ± 1.64e-02	†2.4032e-01 ± 2.18e-02	†2.3968e-01 ± 1.84e-02
9	3.9707e-01 ± 1.27e-02	4.1795e-01 ± 1.96e-02	4.0476e-01 ± 1.02e-02	†4.0201e-01 ± 1.63e-02	4.1117e-01 ± 1.80e-02	4.1667e-01 ± 1.58e-02
10	8.1818e-02 ± 1.46e-02	†9.1375e-02 ± 2.13e-02	1.2401e-01 ± 2.35e-02	9.5804e-02 ± 1.85e-02	9.3706e-02 ± 2.49e-02	†8.8345e-02 ± 1.42e-02
11	1.2866e-01 ± 8.63e-03	†1.3368e-01 ± 1.09e-02	1.4658e-01 ± 1.56e-02	†1.3257e-01 ± 7.42e-03	†1.3271e-01 ± 9.78e-03	†1.2978e-01 ± 6.89e-03
12	3.2878e-01 ± 1.51e-02	3.5564e-01 ± 1.72e-02	3.4125e-01 ± 1.35e-02	4.6115e-01 ± 2.04e-02	4.6259e-01 ± 2.03e-02	4.2278e-01 ± 4.58e-02
13	7.3684e-02 ± 2.33e-02	9.3860e-02 ± 2.99e-02	†8.4211e-02 ± 2.80e-02	†8.4211e-02 ± 2.71e-02	†8.5088e-02 ± 1.92e-02	1.0526e-01 ± 3.01e-02
14	1.2521e-01 ± 1.25e-02	3.1701e-01 ± 1.31e-02	1.6410e-01 ± 3.02e-02	3.2167e-01 ± 1.45e-02	3.2731e-01 ± 1.57e-02	3.2308e-01 ± 1.43e-02
15	1.2457e-01 ± 9.56e-03	1.3526e-01 ± 5.64e-03	1.4124e-01 ± 1.10e-02	1.3319e-01 ± 1.07e-02	1.3590e-01 ± 8.95e-03	1.3348e-01 ± 9.75e-03
16	1.5444e-02 ± 2.43e-03	1.7222e-02 ± 3.40e-03	2.2167e-02 ± 3.94e-03	1.6722e-02 ± 2.98e-03	1.7278e-02 ± 3.46e-03	†1.6222e-02 ± 2.90e-03
17	6.1333e-02 ± 3.89e-02	6.4000e-01 ± 2.26e-16	6.4000e-01 ± 2.26e-16	6.4000e-01 ± 2.26e-16	6.4000e-01 ± 2.26e-16	6.4000e-01 ± 2.26e-16
18	4.2424e-02 ± 2.91e-02	1.5758e-01 ± 3.32e-02	1.7879e-01 ± 3.95e-02	1.7273e-01 ± 3.85e-02	1.6970e-01 ± 3.14e-02	1.6515e-01 ± 2.80e-02
19	2.3210e-01 ± 5.23e-02	2.5926e-01 ± 1.13e-16	2.5926e-01 ± 1.13e-16	2.5926e-01 ± 1.13e-16	2.5926e-01 ± 1.13e-16	2.5802e-01 ± 6.76e-03
20	5.9091e-02 ± 5.22e-02	1.3939e-01 ± 2.65e-02	1.4242e-01 ± 2.60e-02	1.3939e-01 ± 2.37e-02	1.3485e-01 ± 3.27e-02	1.4697e-01 ± 2.85e-02

the only three losses of DAEA on the 5th, 6th, and 8th datasets actually have no significant performance difference from the corresponding best algorithms (i.e., NSGA-II/SDR, NSGA-II, and MOEA/HD) on these datasets. Moreover, compared with the MCE performance of DAEA measured by the first criterion, that measured by the second criterion, as shown in Table VIII and Table SVI in the supplementary material, presents a much higher significant winning ratio, due to the same reason explained in the last sentence of Section V-A.

As for the FN metric, it can be seen from Table VII and Tables SVII and SVIII in the supplementary material that DAEA can obtain a much smaller number of selected features than the total number of features on each dataset. According to Table VII, DAEA achieves the best performance on 8 datasets (from the 13th to the 20th datasets) and the above-average-level performance on most of the rest datasets. Moreover, according to the average Freidman's rankings of FN in Table IX, DAEA generally performs the second best (only lost to MOEA/D) on all the datasets, compared with the other algorithms. In fact, for classification, minimizing the classification error is usually more important than minimizing the number of selected features, although they are often analyzed together. Therefore, it is concluded that DAEA can generally achieve promising and excellent classification performance for feature selection in classification.

TABLE VII

MEAN FN PERFORMANCE ON TEST DATA, WHILE THE BEST PERFORMANCE IS HIGHLIGHTED IN GRAY AND ANY PERFORMANCE NOT SIGNIFICANTLY DIFFERENT FROM THAT OF DAEA IS PREFIXED BY †

No.	DAEA	NSGA-II	MOEA/D	MOEA/HD	NSGA-II/SDR	PMEA
1	5.5667e+00 ± 8.58e-01	†5.3000e+00 ± 9.88e-01	†5.6333e+00 ± 1.27e+00	†5.2667e+00 ± 8.68e-01	2.7667e+00 ± 4.30e-01	†5.7000e+00 ± 1.15e+00
2	5.0000e+00 ± 1.51e+00	†5.0333e+00 ± 1.69e+00	†5.1333e+00 ± 1.89e+00	4.1667e+00 ± 1.46e+00	4.1000e+00 ± 1.56e+00	†4.7667e+00 ± 1.70e+00
3	5.5333e+00 ± 2.11e+00	†4.7333e+00 ± 1.82e+00	4.8000e+00 ± 3.07e+00	4.5000e+00 ± 1.72e+00	2.1333e+00 ± 3.46e-01	4.4000e+00 ± 1.65e+00
4	3.0667e+00 ± 3.65e-01	†3.2000e+00 ± 8.05e-01	†3.3333e+00 ± 1.47e+00	†3.2000e+00 ± 7.61e-01	2.7333e+00 ± 5.21e-01	†3.3333e+00 ± 1.27e+00
5	5.9667e+00 ± 3.53e+00	†4.6667e+00 ± 2.55e+00	4.0667e+00 ± 2.52e+00	†4.3667e+00 ± 2.09e+00	†4.9333e+00 ± 3.20e+00	†5.6333e+00 ± 3.41e+00
6	7.0333e+00 ± 4.08e+00	†6.5667e+00 ± 3.34e+00	†6.1333e+00 ± 4.06e+00	†6.2000e+00 ± 3.56e+00	†5.8000e+00 ± 2.67e+00	†7.3000e+00 ± 3.33e+00
7	1.5333e+01 ± 2.70e+00	1.1733e+01 ± 2.18e+00	1.3600e+01 ± 2.85e+00	9.7333e+00 ± 4.97e+00	8.0333e+00 ± 2.30e+00	1.8033e+01 ± 3.40e+00
8	1.0867e+01 ± 2.75e+00	†1.0600e+01 ± 3.23e+00	8.7333e+00 ± 2.95e+00	†1.0633e+01 ± 4.41e+00	†1.1133e+01 ± 3.94e+00	†1.0733e+01 ± 2.39e+00
9	3.8667e+00 ± 2.60e+00	8.2000e+00 ± 4.89e+00	†2.9667e+00 ± 1.99e+00	†5.3667e+00 ± 3.88e+00	8.5000e+00 ± 4.13e+00	9.4667e+00 ± 3.93e+00
10	2.9967e+01 ± 1.12e+01	†3.0500e+01 ± 8.04e+00	1.3967e+01 ± 5.49e+00	†2.5733e+01 ± 8.19e+00	†3.1267e+01 ± 8.14e+00	†2.9200e+01 ± 8.80e+00
11	9.3167e+01 ± 1.59e+01	†9.0633e+01 ± 1.20e+01	†8.2700e+01 ± 2.42e+01	†8.9767e+01 ± 8.99e+00	8.6167e+01 ± 1.20e+01	†9.0333e+01 ± 1.19e+01
12	1.0567e+01 ± 5.05e+00	1.3500e+01 ± 5.06e+00	†9.3667e+00 ± 5.47e+00	4.3833e+01 ± 3.53e+00	7.2200e+01 ± 6.03e+00	3.5667e+01 ± 1.10e+01
13	9.7000e+00 ± 7.52e+00	3.5033e+01 ± 8.20e+00	1.2100e+01 ± 4.27e+00	3.0500e+01 ± 7.53e+00	5.2133e+01 ± 1.12e+01	4.2300e+01 ± 6.54e+00
14	1.6900e+01 ± 3.41e+00	1.0807e+02 ± 1.09e+01	†1.9867e+01 ± 6.49e+00	1.2437e+02 ± 1.16e+01	1.2120e+02 ± 9.53e+00	1.1590e+02 ± 7.41e+00
15	5.1200e+01 ± 1.35e+01	1.5963e+02 ± 1.40e+01	7.7100e+01 ± 1.84e+01	1.7857e+02 ± 1.68e+01	1.7940e+02 ± 1.60e+01	1.7740e+02 ± 1.50e+01
16	4.1533e+01 ± 4.31e+01	1.5580e+02 ± 1.77e+01	7.7600e+01 ± 3.79e+01	1.6310e+02 ± 1.57e+01	1.8410e+02 ± 1.28e+01	1.6407e+02 ± 1.10e+01
17	5.0333e+00 ± 1.94e+00	7.4927e+02 ± 1.21e+01	3.8233e+02 ± 1.18e+01	9.6897e+02 ± 1.04e+01	9.6747e+02 ± 1.52e+01	8.0297e+02 ± 1.43e+01
18	2.5000e+00 ± 9.38e-01	2.1649e+03 ± 5.80e+01	1.7072e+03 ± 3.42e+01	2.2637e+03 ± 3.35e+01	2.2846e+03 ± 3.91e+01	2.1552e+03 ± 5.53e+01
19	4.1000e+00 ± 2.95e+00	2.4088e+03 ± 2.64e+01	1.9323e+03 ± 3.97e+01	2.5650e+03 ± 7.61e+01	2.6258e+03 ± 2.86e+01	2.4043e+03 ± 2.92e+01
20	1.5667e+00 ± 7.28e-01	2.9333e+03 ± 3.18e+01	2.5578e+03 ± 2.12e+02	3.1178e+03 ± 4.82e+01	3.1632e+03 ± 4.16e+01	2.9346e+03 ± 3.09e+01

TABLE VIII

WIN, LOSS, AND DRAW RATIOS FOR DAEA AGAINST ALL THE OTHER ALGORITHMS ON ALL DATASETS, WHILE VALUES IN BRACKET DENOTE SIGNIFICANT RESULTS OR NO SIGNIFICANT DIFFERENCES

Metric	Data	Win	Loss	Draw
MCE	Train	100% (91%)	0% (0%)	0% (9%)
	Test	92% (62%)	8% (0%)	0% (38%)
	Test*	98% (82%)	2% (0%)	0% (18%)
FN	Train	56% (44%)	43% (27%)	1% (29%)
	Test	61% (47%)	39% (16%)	0% (37%)
	Test*	50% (45%)	48% (31%)	2% (24%)

TABLE IX
AVERAGE FREIDMAN'S RANKINGS ON ALL DATASETS

Metric	Data	DAEA	NSGA-II	MOEA/D	MOEA/HD	NSGA-II/SDR	PMEA
MCE	Train	1.45	3.23	4.34	3.84	4.56	3.57
	Test	2.37	3.45	3.78	3.60	4.06	3.74
	Test*	1.95	3.61	3.99	3.73	3.91	3.82
FN	Train	2.88	3.69	2.51	4.04	3.83	4.05
	Test	2.67	3.56	2.57	4.02	4.09	4.09
	Test*	2.92	3.52	2.46	3.90	4.17	4.03

D. Major Component Contribution Analysis

We have examined the contributions of the three major components in DAEA: 1) the duplication analysis method; 2) the diversity-based selection method; and 3) the reproduction method (including the mating process). First, a baseline algorithm (BA) is made by replacing the tournament selection

TABLE X
MEAN HV PERFORMANCE ON TEST DATA, WHILE ANY PERFORMANCE
BETTER THAN OR NOT SIGNIFICANTLY DIFFERENT FROM THAT OF BA IS
HIGHLIGHTED IN GRAY OR PREFIXED BY †, RESPECTIVELY

No.	BA	BA-dup	BA-div	BA-rep	DAEA
1	8.6902e-01 ± 2.00e-02	†8.6483e-01 ± 2.32e-02	†8.7465e-01 ± 1.84e-02	†8.6946e-01 ± 1.49e-02	†8.7364e-01 ± 1.52e-02
2	8.8772e-01 ± 7.63e-03	†8.8851e-01 ± 6.83e-03	†8.8658e-01 ± 6.78e-03	†8.9063e-01 ± 6.83e-03	†8.9084e-01 ± 7.11e-03
3	9.3339e-01 ± 1.44e-02	†9.3714e-01 ± 3.92e-03	†9.3747e-01 ± 4.80e-03	†9.3876e-01 ± 4.93e-03	†9.3880e-01 ± 4.59e-03
4	8.7655e-01 ± 1.21e-02	†8.7444e-01 ± 1.47e-02	†8.7733e-01 ± 1.30e-02	8.8326e-01 ± 9.16e-03	8.8436e-01 ± 8.83e-03
5	7.8840e-01 ± 2.26e-02	†7.9238e-01 ± 1.56e-02	†7.9245e-01 ± 1.72e-02	†7.9377e-01 ± 1.82e-02	†7.9258e-01 ± 1.96e-02
6	7.7484e-01 ± 2.74e-02	7.9683e-01 ± 1.89e-02	7.9554e-01 ± 2.14e-02	8.0022e-01 ± 1.98e-02	8.0493e-01 ± 2.22e-02
7	6.8097e-01 ± 4.68e-02	†6.9900e-01 ± 2.88e-02	†6.9973e-01 ± 3.44e-02	7.2181e-01 ± 1.57e-02	7.3839e-01 ± 1.44e-02
8	7.1945e-01 ± 1.87e-02	7.5127e-01 ± 1.29e-02	7.4890e-01 ± 1.61e-02	7.5159e-01 ± 1.34e-02	7.5487e-01 ± 1.53e-02
9	5.8343e-01 ± 1.96e-02	6.3135e-01 ± 1.26e-02	6.2435e-01 ± 1.30e-02	6.2705e-01 ± 8.48e-03	6.3221e-01 ± 1.10e-02
10	8.0692e-01 ± 2.29e-02	8.6636e-01 ± 1.77e-02	8.5611e-01 ± 2.18e-02	9.0875e-01 ± 1.22e-02	9.1176e-01 ± 1.19e-02
11	6.9058e-01 ± 1.43e-02	7.4961e-01 ± 1.39e-02	7.4826e-01 ± 1.16e-02	8.3607e-01 ± 9.52e-03	8.3136e-01 ± 9.54e-03
12	6.1155e-01 ± 2.28e-02	†6.1690e-01 ± 3.04e-02	†6.1520e-01 ± 3.40e-02	6.9545e-01 ± 1.51e-02	6.9761e-01 ± 1.34e-02
13	7.7219e-01 ± 2.86e-02	8.2713e-01 ± 2.21e-02	8.1814e-01 ± 2.71e-02	9.1989e-01 ± 2.24e-02	9.2841e-01 ± 2.07e-02
14	5.3310e-01 ± 1.27e-02	5.7963e-01 ± 9.30e-03	5.7524e-01 ± 1.31e-02	8.8125e-01 ± 1.31e-02	8.6580e-01 ± 1.71e-02
15	6.5731e-01 ± 1.21e-02	6.8705e-01 ± 1.24e-02	6.8576e-01 ± 1.11e-02	8.3077e-01 ± 8.56e-03	8.7795e-01 ± 8.11e-03
16	7.4293e-01 ± 1.36e-02	7.6806e-01 ± 9.96e-03	7.7063e-01 ± 1.03e-02	9.1747e-01 ± 7.82e-03	9.8291e-01 ± 2.24e-03
17	2.8636e-01 ± 2.47e-03	2.8448e-01 ± 2.22e-03	2.8479e-01 ± 2.52e-03	5.5216e-01 ± 1.72e-01	9.4363e-01 ± 3.54e-02
18	5.3479e-01 ± 1.70e-02	5.2704e-01 ± 1.72e-02	†5.2962e-01 ± 1.76e-02	6.4032e-01 ± 2.29e-02	9.6124e-01 ± 2.64e-02
19	4.7926e-01 ± 3.55e-03	4.6112e-01 ± 3.96e-03	4.7550e-01 ± 2.50e-03	5.6384e-01 ± 4.58e-03	7.8884e-01 ± 4.75e-02
20	5.3823e-01 ± 2.45e-02	5.3159e-01 ± 1.15e-02	†5.3566e-01 ± 1.73e-02	6.4260e-01 ± 1.90e-02	9.4615e-01 ± 4.75e-02

TABLE XI
WIN AND LOSS RATIOS FOR EACH BA VARIANT AGAINST BA ON ALL
DATASETS, WHILE VALUES IN BRACKET DENOTE SIGNIFICANT RESULTS

Data	Metric	BA-dup	BA-div	BA-rep
Training	Win	70% (55%)	70% (55%)	95% (85%)
	Loss	30% (20%)	30% (15%)	5% (0)
Test	Win	70% (45%)	75% (45%)	100% (80%)
	Loss	30% (20%)	25% (10%)	0 (0)
Test*	Win	80% (55%)	80% (60%)	100% (80%)
	Loss	20% (10%)	20% (10%)	0 (0)

TABLE XII
AVERAGE FREIDMAN'S RANKINGS ON ALL DATASETS

Metric	Data	BA	BA-dup	BA-div	BA-rep	DAEA
HV	Train	4.15	3.78	3.61	2.09	1.38
	Test	4.08	3.53	3.42	2.22	1.75
	Test*	4.19	3.38	3.35	2.41	1.67

method of NSGA-II with a random selection method, while all the other parts of NSGA-II are retained. Then, the aforementioned components of DAEA are, respectively, added into or replacing the similar components of BA, making three variant algorithms, named *BA-dup*, *BA-div*, and *BA-rep*. To be more specific, *BA-dup* is made by adding line 11 of Algorithm 1 and the whole Algorithm 2 into BA; *BA-div* is made by adding

line 11 of Algorithm 1 into BA and also replacing the traditional crowding distance-based selection method with the whole Algorithm 3; and *BA-rep* is made by replacing both mating and variation parts of BA with the whole Algorithm 4. It should be noted both the duplication analysis and diversity maintenance of DAEA need to be connected with the removal of duplicated solutions in the decision space, because analyzing two completely the same solutions makes no sense.

The detailed HV performance of BA and its three variants, along with DAEA, on the test data measured by the first criterion is shown in Table X, while that measured by the second criterion is shown in Table SX in the supplementary material. Their HV performance on the training data is shown in Table SIX in the supplementary material. For the readers' convenience, we also calculate the general winning ratios for each BA variant against BA in Table XI and the average Freidman's rankings in Table XII. The results clearly show that each BA variant performs generally better than BA, with the winning ratios considerably larger than the loss ratios, on both the training and test data. It is also suggested that the reproduction component has the largest contribution, while the other two components have approximately the same contribution. To be more specific, as shown in Table X, *BA-rep* performs better than BA on all the datasets, while *BA-dup* and *BA-div* perform better than BA on most of the datasets but both loss to BA on the high-dimensional datasets (with the number of selected features larger than 2000). This is because both *BA-dup* and *BA-div* attempt to improve the population diversity in the objective space, which in turn will impact the algorithm's convergence ability to some extent. Nevertheless, DAEA still performs better than BA on all the datasets, and performs better than each BA variant on most of the datasets. Thus, it is suggested that the three major components of DAEA can independently make contributions to the algorithm performance and will generally make greater contributions if combined together.

VI. CONCLUSION

The goal of this article was to develop a new method for biobjective feature selection based on duplication analysis. The goal has been successfully achieved by analyzing and handling the duplication in both the decision space and the objective space, proposing a diversity mechanism for feature selection, and modifying the reproduction process of the EA. The new method (DAEA) was examined and compared with five state-of-the-art algorithms on 20 benchmark classification datasets with a variety number of features ranging from 13 to over 7070. The results show that DAEA outperformed all the competing algorithms in terms of both HV and IGD, and successfully selected a much smaller number of features while achieved better classification accuracies on most of these datasets. In addition, DAEA also achieved better convergence and diversity than the competing algorithms. Further analyses of the results revealed that all the three new components (the duplication analysis and handling, the diversity-based selection, and the modified reproduction method) each contributed positively to the success of the new algorithm DAEA for the

good performance, but combining them together into DAEA contributed the most to the final best results.

In the future, we plan to further analyze the parameter sensitivities of DAEA, and study the performance of DAEA on more classification datasets, especially under the multimodal evolutionary environment.

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